HACKCANTON SEASON #2 · GRAND FINAL · 5 AUGUST 2026

tirai.

The confidential multi-dealer RFQ / OTC desk, built native on Canton.
Sealed quotes, atomic delivery-versus-payment, post-trade audit.

You whisper quotes. The market hears nothing.

PUGAR HUDA MANTORO · SOLO · TIRAI.VERCEL.APP · GITHUB.COM/PUGARHUDA/TIRAI

Asking for a price is the information

A bank wants to sell fifty million of bonds. Before it trades, it must ask dealers for a price.

**The moment anyone sees you asking,
they know your size and your direction.**

THE TWO THINGS INSTITUTIONS DO TODAY

Privacy, or proof. Never both.

TRADE IN PUBLIC

Everything leaks

On a transparent chain your request is a transaction and every competing quote is a transaction. Rivals simply read them. Front-running and adverse selection are structural, not incidental.

TRADE ON THE PHONE

Nothing is provable

A voice broker or a chat room is private — and leaves no record. Six months later you cannot prove to compliance that you got the best price available. Nothing is atomic.

Which is why block trading in 2026 still happens on the telephone.

TIRAI GIVES YOU BOTH

The dealer terminal, on the ledger

01 · RFQ	02 · SEALED QUOTE	03 · ESCROW	04 · AWARD	05 · SETTLE
Request, unseen	**Whispered**	**Committed**	**Second price**	**Atomic DvP**
A buyer asks a chosen dealer panel. The market never sees the enquiry.	Each dealer answers the buyer alone. Rivals never receive the contract.	Quoting locks the dealer's bond. A price is a commitment, not a bluff.	Cheapest ask wins and is paid the runner-up's price. Honest quoting dominates.	Bond and cash move in one transaction. The regulator gets the report.

Three execution rails — **reverse-Vickrey auction**, **direct bilateral OTC**, **partial fills** on both. Multi-instrument baskets settle as one atomic multi-leg trade.

You sign in as **one identity** and the home screen is the **book**: every request that party is allowed to see, and nothing else.

LOSING QUOTES ARE ARCHIVED — NEVER REVEALED TO ANYONE

NOT HIDDEN BY THE INTERFACE

Dealer B's node never received dealer A's quote

On Canton a contract is delivered only to its signatories and observers. There is no filter to misconfigure and nothing to decrypt — the data never travels.

ENFORCED BY THE LEDGER

Not by policy

A sealed quote is signed dealer + buyer, with no other observers. The regulator observes executed trades only: zero pre-trade visibility, full post-trade record.

FALSIFIABLE, AND FALSIFIED

Live privacy verifier

A script recomputes each party's visible contract set on the real DevNet ledger and exits non-zero if a single quote ever leaks. It is a test, not a dashboard.

SIGNED IN AS DEALER A, THE ASK IS THERE · SIGNED IN AS DEALER B, THERE IS NOTHING TO HIDE

THE SUPERPOWER

Fifth build. First one the chain did for me.

BUILD	CHAIN	PRIVACY MACHINERY I HAD TO WRITE
Diam	Arbitrum (iExec)	TEE confidential compute, encrypted handles
Segel	Stellar (Soroban)	Two Groth16 circuits, hand-rolled Poseidon
Sealed Pair	Sui	Walrus commitments + Seal threshold encryption
Samar	Ethereum (Zama)	FHE, branchless encrypted settlement
Tirai	Canton	**None. Sub-transaction privacy is the ledger model.**

"Dealer B cannot see dealer A's quote" is a signatory / observer declaration.

WHO BUYS IT

Desks whose tickets are big enough to leak

PRIMARY

Trading desks

Fixed-income and crypto-asset desks at banks, asset managers and prop shops moving $1m–$100m tickets that cannot afford to signal.

CHANNEL

Hosting venues

Temple, Bron, Console, Canton Loop — they hold the client relationship and the custody; Tirai ships as an embedded app and shares the fee.

MANDATORY THIRD PARTY

The regulator

Not a payer, but the reason a private chat will not do. Post-trade-only visibility is the middle ground supervisors ask for.

Below a million, leakage is a rounding error.
Above it, leakage is the whole cost.

HOW IT MAKES MONEY

A venue fee the contract collects itself

PER-TRADE FEE

Basis points of notional

Taken in the settlement asset, atomically inside the settlement transaction. If the trade settles, the fee is paid — no invoicing, no collection risk. The rate is not set: it comes out of a design-partner conversation, not out of my head.

NETWORK REWARDS

CIP-0047 activity markers

Featured-app markers accrue on every settlement, so a live desk keeps earning from the volume it clears, on top of the fee.

Fee on notional means one desk clearing institutional size is a real business — and tokenised bond issuance is the growth curve underneath it.

ZERO REVENUE TODAY · NO PAYING CUSTOMER · THE MODEL IS DESIGNED, NOT COLLECTED

DIFFERENTIATION

Not a DEX. Not a chat room.

ALTERNATIVE	THE SINGLE AXIS IT FAILS ON
Public order book / on-chain RFQ	Pre-trade confidentiality. The order *is* the signal.
Voice or chat broker	Provable settlement. Nothing is atomic; best execution is reconstructed from chat logs.
Tradeweb / MarketAxess / bank dark pools	The venue itself sees everything. Confidentiality is an operator's promise.
Other Canton privacy demos	No mechanism. They show Canton can keep a secret; they do not run a market.

Private price discovery — with a receipt.

LIVE ON CANTON DEVNET — VERIFIABLE RIGHT NOW

Not a mock-up

2 DEVNET PARTICIPANTS ONE PACKAGE ID	**49** SETTLED TRADES + 5 ATOMIC BASKETS	**16** BEST-EXECUTION ATTESTATIONS	**36** DAML TEST SCRIPTS GREEN

Deployed on HackCanton's own node and the shared 5N validator under the same package id. Hosted read-only desk over live ledger state at **tirai.vercel.app**.

Honest label: that ledger history was generated by me. It demonstrates the product works — it is not customer traction, and I will not present it as such.

THE TWO PROOF VIEWS

The claim is checked against the live ledger

VERIFY PRIVACY

Each party's real visible set

Not a claim — a query. Each dealer sees only its own quotes. The regulator sees zero contracts before settlement. Green, on the network, on demand.

BEST EXECUTION

Proof without an order book

A buyer, or a dealer defending its pricing, discloses a single sealed quote to the regulator. Sixteen attestations prove the clearing price beat every disclosed rival ask.

Confidential pre-trade. Provable post-trade.

YOUR FEEDBACK ASKED FOR REAL EXTERNAL SETTLEMENT

Two assets. Neither of them mine.

The cash leg settles in **real Canton Coin** through the DSO's registry and in **real CBTC** through the DA Utility Registry that BitSafe issues on. Issuers I do not administer and cannot mint into.

CANTON COIN

6 settlements

Four reverse-Vickrey, two direct OTC. 60,900 CC moved to the dealers that won them, each bond-against-cash in one transaction.

CBTC

0.34 + 0.22

A faucet grant accepted on-ledger, then spent through the desk: cleared at the Vickrey second price, and hit directly at the ask.

CETH

One field away

The instrument administrator is the only difference. The rail appears in the desk the day the tokens land, with no release in between.

Nothing in the model is per-asset: `cashInstrument` is any `{admin, id}`, and each registry only differs in where it keeps its off-ledger choice contexts. The desk's Settlement rails view lists every one it can clear on.

THE TEAM

One person. Fifth implementation. Five days.

WHY IT MOVED THIS FAST

The hard part is behind me

Repo created 22 July. The whole CIP-0056 settlement leg landed 23 July. Hosted desk live 24 July. Second participant deployed and privacy-verified 26 July. The product design and the auction mechanics stopped being hypotheses four builds ago.

WHAT I WILL NOT DRESS UP

The honest gaps

No signed design partner and no customer conversation yet. No live cETH or CBTC transaction. DevNet only, no mainnet. No credit or netting layer. One person, so no SLA and no third-party audit.

Since the feedback there is a written 90-day validation plan with the stop criteria set *in advance* — a plan that cannot fail is not a plan.

TIRAI — INDONESIAN FOR "CURTAIN"

Price discovery happens behind it.

I need one design-partner desk. One hour a week for ninety days — not a purchase order.

1 ONE TRADING DESK AS DESIGN PARTNER

2 CETH AND CBTC TEST-TOKEN GRANTS

3 A HOSTING VENUE FOR A SUPERVISED PILOT

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